

Avian Flight Image Classifier: Distinguishing In-Flight Birds Using Deep Learning

github.com/10kshaadow/avian_binary_flight_classifier

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Abstract—Determining whether a bird is in flight from a single static image is a fine-grained visual recognition problem with applications in wildlife monitoring, migration tracking, and aviation safety. This paper presents a binary deep-learning classifier that labels bird images as either *in-flight* or *not in-flight*. We evaluate whether pre-trained convolutional neural networks (CNNs) and Vision Transformers (ViTs) transfer effectively to this task while learning bird-local cues such as wing extension and body pose rather than relying on environmental context. Using transfer learning, targeted data augmentation, and attention-based interpretability analysis, we examine the effects of dataset bias, class imbalance, and shortcut learning. Results show that all models learned the task effectively, with ViT-DINO providing the most consistently bird-centered interpretability, while EfficientNet-B3 benefited the most from the refactored training pipeline. Overall, the findings suggest that transfer learning is effective for binary avian flight classification, but interpretability remains necessary to verify whether strong performance reflects meaningful bird-centered reasoning rather than dataset bias.

I. INTRODUCTION

Automated visual recognition of animal behavior from static images is an active and challenging area of computer vision. Among such behaviors, detecting whether a bird is in flight is a particularly compelling binary classification problem because of its dual relevance to ecological research and public safety. In aviation, bird strikes cost the industry an estimated \$1.2 billion annually in the United States alone [1], and airborne-bird detection is a core component of proposed strike-prevention systems. In ecology, flight behavior serves as a proxy for migration activity, foraging patterns, and population health, data that is traditionally expensive to collect at scale [2].

Despite these motivations, distinguishing in-flight birds from perched or ground-level birds in unconstrained photographs is non-trivial. A model must identify subtle pose cues such as extended wings, an aerodynamic body posture, and feather spread, that may vary significantly across the over 10,000 known avian species. At the same time, models trained on internet-sourced image datasets risk learning spurious correlations: a bird photographed against a blue sky is far more likely to be in-flight than one photographed against a forest

floor, but a robust classifier should rely on the bird itself rather than its background.

This work makes the following contributions:

- We frame avian flight detection as a binary classification task and curate a labeled dataset drawn from publicly available wildlife image repositories.
- We benchmark multiple pre-trained architectures including ResNet-50 [3], EfficientNet [4], and Vision Transformer (ViT) [5] under different fine-tuning regimes.
- We analyze attention and gradient maps to assess whether models attend to anatomically relevant regions.
- We discuss limitations including species imbalance, pose ambiguity near takeoff/landing, and background leakage.

II. RELATED WORK

A. Fine-Grained Visual Recognition

Fine-grained visual recognition (FGVR) seeks to distinguish subcategories within a broader class e.g., bird species, car models, or aircraft variants. The CUB-200-2011 dataset [6] has long served as the standard benchmark for bird species classification, and methods such as Bilinear CNNs [7] and part-based attention networks demonstrated that capturing local discriminative regions is critical for high accuracy. Our task differs from species classification in that it is behavioral rather than taxonomic, but the core challenge of identifying fine-grained pose features over background clutter is shared.

B. Transfer Learning for Wildlife Imagery

Pre-training on large datasets such as ImageNet [8] and then fine-tuning on domain-specific data has become the dominant paradigm for wildlife image analysis. Norouzzadeh et al. [9] demonstrated that deep CNNs pre-trained on ImageNet could classify animals in camera-trap images across dozens of species with high accuracy, providing strong evidence that learned visual representations transfer well to wildlife domains. Beery et al. [10] further highlighted the difficulty of generalizing to novel environments, a challenge directly relevant to our concern about background spurious correlations.

C. Vision Transformers

Dosovitskiy et al. [5] introduced the Vision Transformer (ViT), which applies the self-attention mechanism of Transformer architectures [11] to sequences of image patches. ViTs have since achieved state-of-the-art performance on numerous image classification benchmarks, often matching or exceeding CNNs when sufficient data or pre-training is available. Importantly for our task, the attention maps of ViTs offer an interpretability advantage: attention weights can be visualized to inspect which image regions the model focuses on, providing a natural mechanism for assessing whether flight-relevant anatomy rather than background drives predictions [12].

D. Shortcut Learning and Spurious Correlations

A well-documented failure mode of deep learning classifiers is shortcut learning, where models exploit statistical correlations in training data that do not reflect causal features [13]. In wildlife datasets, background environment is a classic shortcut: a classifier may learn “sky $\Rightarrow$ flying” rather than “spread wings $\Rightarrow$ flying.” Techniques to mitigate this include background augmentation, foreground segmentation, and domain-adversarial training [14]. Our proposed solution incorporates background augmentation as a first-line defense.

III. PROPOSED SOLUTION

A. Dataset Construction

We source images from iNaturalist, which provides community-verified behavioral metadata. Images are labeled *in-flight* if the bird is fully airborne with wings extended, and *not in-flight* otherwise (perched, walking, swimming). We apply stratified splitting into 70% train, 15% validation, and 15% test sets, balancing classes to mitigate the natural species-level imbalance documented in the literature [15].

B. Model Architecture

Following the transfer-learning paradigm validated by Norouzzadeh et al. [9] and Tan and Le [4], we evaluate three backbone architectures:

- 1) **ResNet-50** [3]: A residual CNN pre-trained on ImageNet-1k. The final fully connected layer is replaced with a two-class head. We experiment with both frozen-backbone (linear probe) and full fine-tuning regimes.
- 2) **EfficientNet-B3** [4]: A compound-scaled CNN offering improved parameter efficiency. Prior work has shown strong transfer performance on fine-grained wildlife tasks with relatively small datasets.
- 3) **ViT-B/16** [5]: A Vision Transformer pre-trained with DINO self-supervised objectives [16], which produces particularly interpretable attention maps useful for diagnosing shortcut learning.

C. Training Strategy

All models are trained with the AdamW optimizer with a cosine annealing learning rate schedule, an initial learning rate of 1×10^{-4} , and weight decay of 1×10^{-2} . We use binary cross-entropy loss with label smoothing ($\epsilon = 0.1$) to reduce

overconfidence. Batch size is set to 32 for CNNs and 16 for ViT due to memory constraints. Early stopping on validation AUC is applied with a patience of 10 epochs.

D. Data Augmentation

To discourage shortcut learning on background cues, we apply the following augmentation pipeline during training, motivated by findings in [13]:

- Random horizontal flips and $\pm 30^\circ$ rotations.
- Color jitter (brightness, contrast, saturation).
- Random background replacement using *Copy-Paste* augmentation [17]: bird foregrounds (isolated via a pre-trained segmentation model) are composited onto randomly sampled indoor and urban background images, forcing the model to rely on the bird rather than its environment.
- CutMix [18] between in-flight and not-in-flight samples to improve decision boundary robustness.

E. Evaluation and Interpretability

We report accuracy, AUC-ROC, F1-score, and confusion matrices on the held-out test set. To assess whether models attend to anatomically relevant regions, we compute:

- **Grad-CAM** [19] saliency maps for CNN models, highlighting the spatial regions most influential to the predicted class.
- **Attention rollout** [20] for ViT models, aggregating attention across all heads and layers.

Maps will be qualitatively evaluated by domain experts to confirm whether attention concentrates on wings and body posture versus sky or ground background. A quantitative evaluation using IoU against manually drawn wing-region masks is planned for the final version.

F. Expected Outcomes

We hypothesize that ViT-B/16 with background augmentation will outperform ResNet-50 in both accuracy and interpretability, consistent with findings that self-attention captures global structural relationships more effectively than local convolutional filters [5], [16]. We further expect that background augmentation will improve generalization across geographically diverse test images, reducing the gap between in-distribution and out-of-distribution performance documented by Beery et al. [10].

IV. ACTUAL SOLUTION

The actual implementation diverged from the proposed design in several ways.

- **Preprocessing - YOLO-Based Bird Localization:** The most significant addition not present in the proposal is a dedicated YOLO preprocessing stage (Section 3 of the notebook). YOLOv8n is applied to every image before any augmentation or training, detecting the bird using COCO class ID 14 and cropping a padded bounding box (15% padding fraction) around it. Images where YOLO fails to detect a bird above a 0.20 confidence threshold

fall back to the original uncropped image. This stage directly replaces the proposed Copy-Paste background replacement augmentation: rather than compositing bird foregrounds onto synthetic backgrounds, the pipeline eliminates background context upstream by cropping to the bird itself. The segmentation-based foreground isolation described in the proposed solution was not implemented.

- **Augmentation Pipeline:** CutMix, which was proposed as a technique to improve decision boundary robustness between classes, was not included in the final training transforms. The actual augmentation pipeline consists of RandomResizedCrop (scale 0.6–1.0), RandomHorizontalFlip, RandomRotation($\pm 30^\circ$), ColorJitter, RandomGrayscale ($p=0.05$), and RandomErasing ($p=0.25$, scale 0.02–0.12). RandomErasing was an addition made to the proposed solution; it randomly occludes rectangular patches of the input tensor after normalization, serving a similar regularization purpose to CutMix without requiring cross-sample mixing.
- **Loss Function and Class Imbalance Handling:** The proposed solution specified binary cross-entropy loss, whereas the final implementation uses `nn.CrossEntropyLoss` with label smoothing ($\epsilon = 0.1$), which is functionally equivalent for a two-class head with softmax outputs. This implementation adds inverse-frequency class weighting to the criterion based on the training set class counts which addresses potential class imbalance in the dataset.
- **Training Hyperparameters:** Early stopping patience was reduced from 10 epochs to 7. The maximum epoch count was set to 20. The ViT batch size was not differentiated from the CNN batch size. All models use a uniform batch size of 32, rather than the proposed 16 for ViT due to memory constraints. All three backbones are trained with full fine-tuning; the frozen-backbone linear probe variant described for ResNet-50 was not explored. The optimizer (AdamW), learning rate ($1E-4$), weight decay ($1E-2$), and cosine annealing schedule remain as proposed.
- **Interpretability:** Both Grad-CAM (for ResNet-50 and EfficientNet-B3) and attention rollout (for ViT-DINO) were implemented as originally specified. However, the planned IoU evaluation against manually drawn wing masks was not included and remains future work. Interpretability was therefore assessed only through qualitative heatmap inspection.

V. RESULTS

A. Proposed Solution

All three models showed a clear downward trend in training loss, indicating that they learned the binary task effectively. As shown in Fig. 1, the CNN models were subjected to early stopping after validation AUC-ROC plateaued for several consecutive epochs, suggesting relatively fast convergence under transfer learning. ResNet-50 stabilized earlier, while

EfficientNet-B3 improved more gradually. In contrast, ViT-DINO maintained strong validation performance with somewhat greater fluctuation across epochs, reflecting a different fine-tuning dynamic.

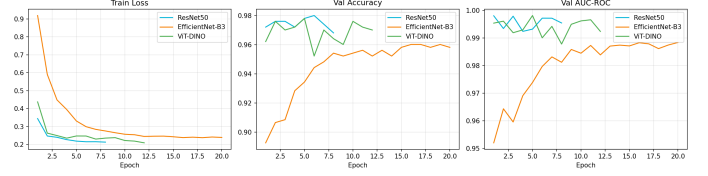


Fig. 1. Proposed solution training results.

While scalar metrics summarized overall performance, they did not reveal the structure of each model’s mistakes. Therefore, the confusion matrices in Fig. 2 were used to analyze class-specific prediction behavior, showing how often *in-flight* and *not-in-flight* images were correctly classified and where misclassifications occurred across the three architectures.

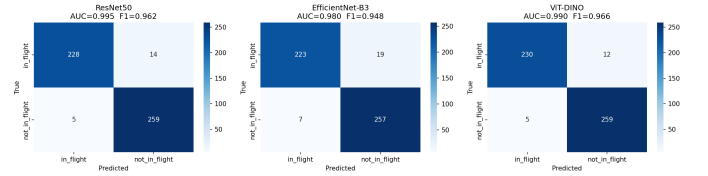


Fig. 2. Proposed solution confusion matrices.

Fig. 2 showed that both ResNet-50 and ViT-DINO achieved strong class separability and balanced classification performance, as reflected by their high AUC-ROC and F1 scores: ResNet-50 (AUC = 0.995, F1 = 0.962) and ViT-DINO (AUC = 0.990, F1 = 0.966). Their confusion matrices also exhibited very similar prediction distributions, suggesting that the two models made errors in broadly comparable ways. EfficientNet-B3 (AUC = 0.980, F1 = 0.948) underperformed, compared to its two counterparts because it was skewed towards falsely classifying *not-in-flight* images as *in-flight* by a slight margin.

Afterwards, it became evident that three different pretrained models were learning to classify the images correctly. A more in-depth interpretability analysis was then carried out using Grad-CAM saliency maps for the CNNs and attention rollout for the ViT model to determine which weights and biases the models had settled on for classification.

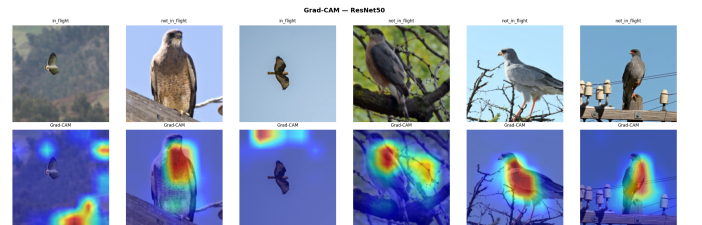


Fig. 3. Proposed solution Grad-CAM, ResNet-50



Fig. 4. Proposed solution Grad-CAM, EfficientNet-B3

The Grad-CAM saliency maps for the CNN models in Figs. 3 and 4 revealed a substantial reliance on contextual information rather than consistently bird-centered reasoning. While some *not-in-flight* examples showed activations aligned with the bird body or nearby perch, many *in-flight* examples were much less convincing. In several cases, the saliency maps focused primarily on surrounding sky or other background regions, with only weak overlap with the bird itself. This was a significant limitation, because it meant that strong classification metrics did not necessarily imply robust bird-centered understanding. Therefore, the saliency analysis indicated that the CNNs were often classifying the images without using the bird itself as the primary indicator.



Fig. 5. Proposed solution Attention rollout, ViT-DINO

The attention rollout results for ViT-DINO were more bird-centered than the CNN saliency maps. In Fig. 5, the highlighted regions were often concentrated on the bird body, wings, or perch area rather than spreading broadly across unrelated background regions. Although some surrounding context still appeared in a few examples, the overall attention patterns aligned more consistently with the bird itself, making the ViT-DINO interpretability results more convincing than those of the CNN models.

Overall, the initial results did not fully support the original hypothesis, suggesting that the proposed solution required further refinement for the improvement of the CNNs.

B. Refinement

To address the clear shortcut-learning tendency toward background cues, the training pipeline was refactored with more aggressive and more targeted data augmentation. First, the random crop range was tightened from (0.6, 1.0) to (0.75, 1.0), which reduced the amount of irrelevant scene context included in each training sample and forced the bird to occupy a larger fraction of the image. Second, color jitter was strengthened and the grayscale probability was increased from 0.05 to 0.125

in order to weaken color-based correlations, such as blue sky or vegetation tones, that could otherwise act as shortcuts for the classifier. In addition, Random Erasing was introduced to mask small image regions during training, making it harder for the models to rely on a single highly predictive patch and encouraging more distributed feature use. Finally, the CNN input size was increased to 300 so that smaller birds, especially airborne ones, retained more visible detail after resizing, since CNN-based models often benefit from higher input resolutions when learning fine-grained visual features. Collectively, these changes were intended to reduce dependence on spurious background structure and push the models toward learning more bird-local, task-relevant visual features.

C. Overall Findings

The refined training curves in Fig. 6 showed that ResNet-50 and EfficientNet-B3 followed a broadly similar learning behavior, with both models exhibiting steady reductions in training loss and strong validation performance. However, EfficientNet-B3 showed the larger relative improvement compared to its initial run, indicating that it benefited more from the tighter crop range, stronger augmentation, random erasing and higher input resolution. ResNet-50 remained the strongest final CNN overall, with an AUC of 0.993 and F1 of 0.986, while EfficientNet-B3 reached an AUC of 0.991 and F1 of 0.974, as reflected in Fig. 7.

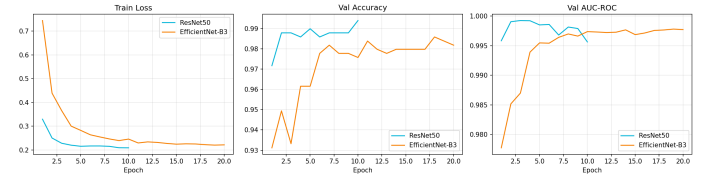


Fig. 6. Refined CNNs training, ResNet-50 & EfficientNet-B3

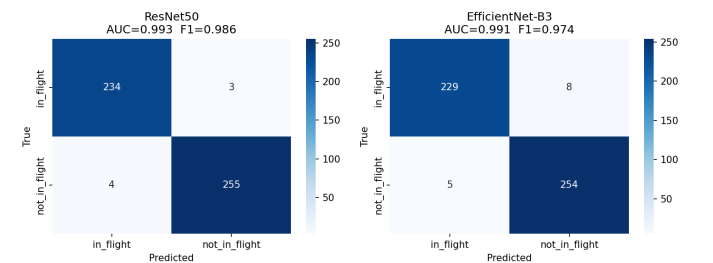


Fig. 7. Refined CNNs training, ResNet-50 & EfficientNet-B3

However, the scalar metrics remained broadly consistent with the proposed solution results. The clearer change appeared in the interpretability analysis, where the refined Grad-CAM visualizations in Figs. 8 and 9 provided the strongest evidence of improvement.

The refined ResNet-50 Grad-CAM results in Fig. 8 did not show a meaningful overall improvement in interpretability, as the model continued to rely heavily on surrounding sky, branches, and background texture. In contrast, the

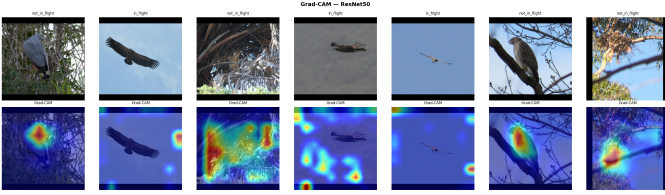


Fig. 8. Refined Grad-CAM, ResNet-50

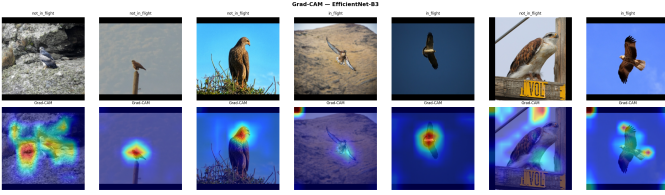


Fig. 9. Refined Grad-CAM, EfficientNet-B3

EfficientNet-B3 Grad-CAM in Fig. 9 showed a more localized distribution of attention around the bird region, particularly along the wing span and exposed torso during flight. Although some activations still extended beyond the bird, the refined model more clearly traced the bird’s silhouette than in the initial run, indicating more subject-centric attention during classification.

What was consistent across both CNNs was that the *not-in-flight* class relied on fewer obvious shortcuts than the *in-flight* class. In many of these cases, the saliency maps centered on the bird’s closed wing posture and resting body configuration, although some close-up perched examples still caused the models to incorporate nearby twigs, trunks, or overshadowing structures into the decision. Overall, the refactored pipeline appeared to have a limited effect on ResNet-50 but a more noticeable impact on EfficientNet-B3. For the transformer model, the ViT-DINO attention rollout in Fig. 10 remained largely unchanged from the initial run. Its attention maps continued to align reasonably well with the bird region, and its scalar performance remained similarly strong. As a result, the main interpretability changes after refactoring were concentrated in the CNN models, particularly EfficientNet-B3.



Fig. 10. Refined attention rollout, ViT-DINO

VI. FUTURE WORKS

Several directions remain open for extending this system in future work.

First, the current dataset likely underrepresented boundary cases of the classification task (ex. perched birds silhouetted against open sky, or birds in flight against dense, cluttered backgrounds). Augmenting the training and evaluation sets with such hard negatives and hard positives would stress-test the model’s reliance on genuine postural and morphological cues rather than coarse scene-level statistics, and would provide a more realistic estimate of out-of-distribution performance.

Second, the present work treats flight classification as a static, per-image decision. Many real-world applications—behavioral monitoring, wildlife survey automation, and ecological event detection—require understanding flight as a temporal phenomenon. Extending the pipeline to video-based flight-stage recognition, for example by incorporating temporal backbones such as Video Swin Transformer or convolutional LSTM heads over frame sequences, would enable detection of transitional stages such as takeoff and landing that are not well-represented by any single frame.

Third, while the implemented YOLO preprocessing step mitigates background shortcut learning by cropping to the bird prior to classification, it is a heuristic solution that is sensitive to detector confidence thresholds and fails silently when detection is missed. A more principled approach would jointly learn localization and classification. For instance, through instance segmentation models such as SAM or Mask R-CNN—so that the model’s spatial attention is explicitly constrained to the bird’s silhouette during training. Tighter integration between the localization and classification objectives could also enable end-to-end training and open the door to weakly supervised approaches that do not require bounding box annotations.

VII. CONCLUSION

In conclusion, this project showed that binary avian flight classification from single static images was achievable using transfer learning with both CNN and transformer-based architectures. All three pretrained models learned the task effectively, with ResNet-50 and ViT-DINO achieving the strongest initial results. However, interpretability analysis revealed that high scalar performance did not always correspond to bird-centered reasoning, as the CNN saliency maps often relied on contextual shortcuts such as sky, branches, and surrounding background, while the ViT-DINO attention rollout remained more consistently aligned with the bird itself.

The refactored training pipeline introduced tighter crop ranges, stronger color and grayscale augmentation, Random Erasing, and larger CNN input resolution to reduce shortcut learning. These changes had limited effect on ResNet-50 but produced a more noticeable improvement for EfficientNet-B3, whose Grad-CAM maps showed more subject-centric attention than in the initial run. Overall, the results suggest that transfer learning can solve the task effectively, but interpretability remains essential for determining whether high-performing models are learning meaningful bird-local cues rather than exploiting dataset bias.

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